

# Complete Guide: How to Run Python Code in Visual Studio Code

Windows — From Python installation to running and debugging a Python program

## 1. Install Python

Download Python from <https://www.python.org/downloads/>. Run the installer. IMPORTANT: check “Add python.exe to PATH” before clicking “Install Now”.

### Verify Python

Open Command Prompt (Windows + R → type cmd → Enter) and run:

```
python --version
```

You should see something similar to:

```
Python 3.x.x
```

You can also check:

```
py --version
```

## 2. Install Visual Studio Code

Download VS Code from <https://code.visualstudio.com/>. Install it using the normal/default options, then open Visual Studio Code.

## 3. Install the Python Extension

Open VS Code → Extensions, or press Ctrl + Shift + X. Search for “Python” and install “Python” by Microsoft.

## 4. Create a Python Project Folder

```
Documents
├── Python Projects
│   └── First Project
```

## 5. Open the Project in VS Code

Select File → Open Folder, choose your project folder, and click Select Folder.

## 6. Create Your First Python File

Create a new file named:

```
hello.py
```

The .py extension is used for Python files.

## 7. Write Your First Python Program

```
print("Hello World")
```

Save with Ctrl + S.

## 8. Select the Python Interpreter

Press Ctrl + Shift + P → search “Python: Select Interpreter” → select your installed Python version.

## 9. Run the Python Program

### Method 1 — Run Button

Open hello.py and click  Run Python File in the top-right corner.

### Method 2 — Terminal

Open Terminal → New Terminal, or press Ctrl + ` . Then run:

```
python hello.py
```

Expected output:

```
Hello World
```

## 10. If the python Command Does Not Work

Try the Python Launcher:

```
py hello.py
```

Check it with:

```
py --version
```

## 11. If Neither python nor py Works

Check whether Windows can find Python:

```
where python
where py
```

If Python is installed but cannot be found, the PATH environment variable may need to be configured.

## 12. Add Python to PATH Manually

A typical Python installation may be located at:

```
C:\Users\YourName\AppData\Local\Programs\Python\Python313\
```

Open Windows Search → Environment Variables → Edit the system environment variables → Environment Variables. Under User variables, select Path → Edit → New.

Add the Python folder and Scripts folder:

```
C:\Users\YourName\AppData\Local\Programs\Python\Python313\
C:\Users\YourName\AppData\Local\Programs\Python\Python313\Scripts\
```

Click OK on all windows, close and reopen VS Code, then run:

```
python --version
```

## 13. Check Which Python VS Code Is Using

```
python --version
where python
```

You can also use Ctrl + Shift + P → Python: Select Interpreter.

## 14. Running a Python Program With User Input

```
name = input("Enter your name: ")  
print("Hello", name)
```

Run:

```
python input.py
```

## 15. Running a Larger Python Project

```
Python Project  
■  
■■■ main.py  
■■■ calculator.py  
■■■ data.txt
```

Open the project folder in VS Code and run:

```
python main.py
```

## 16. Installing Python Packages

```
python -m pip install requests
```

Example:

```
import requests  
print("Requests installed successfully")
```

## 17. Installing Multiple Packages

If the project contains requirements.txt:

```
python -m pip install -r requirements.txt
```

## 18. Create a Virtual Environment

```
python -m venv .venv
```

This creates a .venv folder inside the project.

## 19. Activate the Virtual Environment

### Command Prompt

```
.venv\Scripts\activate
```

### PowerShell

```
.venv\Scripts\Activate.ps1
```

After activation, the terminal usually shows (.venv) at the beginning.

## 20. Select the Virtual Environment

Press Ctrl + Shift + P → Python: Select Interpreter → select the interpreter inside .venv.

## 21. Install Packages Inside the Virtual Environment

```
python -m pip install requests
```

## 22. Run the Project

```
python main.py
```

You can also click  Run Python File.

## 23. Debugging a Python Program

```
x = 10
y = 20

result = x + y

print(result)
```

Click beside a line number to create a breakpoint. Select Run and Debug or press F5. VS Code will pause at the breakpoint so you can inspect variables.

## 24. Common Python Commands

```
python --version
py --version
python filename.py
python -m pip install package_name
python -m pip uninstall package_name
python -m pip list
python -m venv .venv
.venv\Scripts\activate
```

## 25. Common Problems and Solutions

### “python is not recognized”

Try `py --version`. If it works, run `py filename.py`. If neither works, check Python installation and PATH.

### No Python interpreter in VS Code

Press `Ctrl + Shift + P` → Python: Select Interpreter → select your installed Python version.

### Run button is missing

Make sure the Python extension by Microsoft is installed, then restart VS Code.

### pip is not recognized

Use:

```
python -m pip install requests
```

### Wrong Python version

Use Python: Select Interpreter and choose the required version.

## 26. Recommended Basic Workflow

1. Install Python. 2. Verify `python --version`. 3. Install VS Code. 4. Install the Python extension by Microsoft. 5. Create/open your project folder. 6. Create `main.py`. 7. Select the Python interpreter. 8. Write your code. 9. Save with `Ctrl + S`. 10. Run `python main.py` or click  Run Python File.

## 27. Complete First-Test Example

```
name = input("Enter your name: ")
age = int(input("Enter your age: "))

print()
print("Hello", name)
print("You are", age, "years old.")
```

```
print("Python is running successfully in VS Code!")
```

Run:

```
python main.py
```

Example output:

```
Enter your name: ALI
Enter your age: 23

Hello ALI
You are 23 years old.
Python is running successfully in VS Code!
```

## 28. Final Checklist

- Python is installed
- Python was added to PATH
- `python --version` works
- VS Code is installed
- Python extension by Microsoft is installed
- A Python interpreter is selected
- A .py file has been created
- The file has been saved
- The program runs from the VS Code terminal
- Required packages are installed if needed
- A virtual environment is used for larger projects

## The Most Important Command

```
python filename.py
```

For example:

```
python main.py
```

If this successfully runs your program, your basic Python + VS Code setup is working correctly.